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SYSTEMATIC REVIEW

Effects of Periodization on Strength and Muscle Hypertrophy in Volume-Equated Resistance Training Programs: A Systematic Review and Meta-analysis

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Abstract

Background In resistance training, periodization is often used in an attempt to promote development of strength and muscle hypertrophy. However, it remains unclear how resistance training variables are most effectively periodized to maximize gains in strength and muscle hypertrophy.

Objective The aims of this study were to examine the current body of literature to determine whether there is an effect of periodization of training volume and intensity on maximal strength and muscle hypertrophy, and, if so, to determine how these variables are more effectively periodized to promote increases in strength and muscle hypertrophy, when volume is equated between conditions from pre to post intervention.

Methods Systematic searches were conducted in PubMed, Scopus and SPORTDiscus databases. Data from the individual studies were extracted and coded. Meta-analyses using the inverse-variance random effects model were performed to compare 1-repetition maximum (1RM) and muscle hypertrophy outcomes in (a) non-periodized (NP) versus periodized training and (b) in linear periodization (LP) versus undulating periodization (UP). Subgroup analyses examining whether results were affected by training status were performed. Meta-analyses of other periodization model comparisons were not performed, due to a low number of studies.

Results Thirty-five studies met the inclusion criteria. Results of the meta-analyses comparing NP and periodized training demonstrated an overall effect on 1RM strength favoring periodized training (ES 0.31, 95% confidence interval (CI) [0.04, 0.57]; $Z=2.28$, $P=0.02$). In contrast, muscle hypertrophy did not differ between NP and periodized training (ES 0.13, 95% CI [–0.10, 0.36]; $Z=1.10$, $P=0.27$). Results of the meta-analyses comparing LP and UP indicated an overall effect on 1RM favoring UP (ES 0.31, 95% CI [0.02, 0.61]; $Z=2.06$, $P=0.04$). Subgroup analyses indicated an effect on 1RM favoring UP in trained participants (ES 0.61, 95% CI [0.00, 1.22]; $Z=1.97$ ($P=0.05$)), whereas changes in 1RM did not differ between LP and UP in untrained participants (ES 0.06, 95% CI [–0.20, 0.31]; $Z=0.43$ ($P=0.67$)). The meta-analyses showed that muscle hypertrophy did not differ between LP and UP (ES 0.05, 95% CI [–0.20, 0.29]; $Z=0.36$ ($P=0.72$)).

Conclusion The results suggest that when volume is equated between conditions, periodized resistance training has a greater effect on 1RM strength compared to NP resistance training. Also, UP resulted in greater increases in 1RM compared to LP. However, subgroup analyses revealed that this was only the case for trained and not previously untrained individuals, indicating that trained individuals benefit from daily or weekly undulations in volume and intensity, when the aim is maximal strength. Periodization of volume and intensity does not seem to affect muscle hypertrophy in volume-equated pre-post designs. Based on this, we propose that the effects of periodization on maximal strength may instead be related to the neurophysiological adaptations accompanying resistance training.

1 Introduction

During recent decades multiple health-related benefits of resistance training have been documented [1, 2]. Accordingly, resistance training is increasingly acknowledged as a viable exercise modality for improving strength, general fitness, and health-related outcomes [1]. Furthermore, resistance training has been proven highly versatile in the context of sports, leisure, or rehabilitation training [3]. While some health benefits of resistance training emerge independently of increases in muscular strength [1, 2], one of the primary reasons why resistance training is used by practitioners is to

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Key Points

Periodization is the manipulation of training variables over time, and is often used in resistance training programs in an attempt to promote increases in strength and muscle hypertrophy. In order to assess the effect of periodization on strength and muscle hypertrophy, meta-analyses assessing changes from pre- to post-intervention between volume-equated resistance training programs were performed.

When comparing the change from pre- to post-intervention between volume-equated periodized and non-periodized resistance training, periodized resistance training leads to greater increases in strength compared to non-periodized resistance training.

Undulating periodization leads to greater increases in strength than linear periodization in volume-equated pre-post designs. However, this was only observed for resistance trained individuals, and not for untrained individuals.

Resistance training periodization does not appear to affect muscle hypertrophy when volume is equated between conditions. This indicates that other adaptations accompanying resistance training—for example, neurophysiological changes—may explain the observed positive effects of resistance training periodization on strength gains.

improve muscular strength and promote muscle hypertrophy. Therefore, discovering how to maximize muscle hypertrophy and muscular strength is of great interest to a wide range of individuals including athletes, patients, trainers, and clinical therapists. While resistance training represents an efficient mean of increasing both strength and muscle mass, it remains to be fully determined how to optimize resistance training protocols to maximize these measures.

Variables that can be manipulated to improve the strength and muscle hypertrophy responses to resistance training include, but are not limited to, training intensity, training volume, and training frequency. The current body of literature provides empirical evidence regarding the influence of these variables on development of strength and muscle hypertrophy [4–10]. However, it remains unclear how manipulation of these variables over time affects strength gains and muscle hypertrophy.

The manipulation of training variables over time is known as periodization. In sports such as powerlifting and weightlifting, which are highly dependent on strength and where athletes only compete a few times a year, periodization is used with the aim of maximizing performance specifically on the day of competition. For many years this practice was largely based on the rationale that by adhering to the principles of the general adaptation syndrome theory, training could be planned in a manner where overtraining and injury could be avoided, and performance could be maximized [11]. In the current body of literature, different periodization models have been implemented including linear periodization (LP), reverse linear periodization (RLP), undulating periodization (UP), and block periodization (BP). LP, also termed traditional periodization, is characterized by increases in intensity and decreases in volume over time. RLP is characterized by decreases in intensity and increases in volume over time. UP, also termed non-linear periodization, is characterized by more frequent variations in volume and intensity than LP and RLP, and can be divided into two subcategories: daily undulating periodization (DUP), characterized by daily undulations in intensity and volume, and weekly undulating periodization (WUP), characterized by weekly undulations in intensity and volume. In BP, training is divided into several distinct blocks, where each training block is structured to promote the development of a specific training goal. An example of this is a block in which the development of muscle hypertrophy is the main aim, followed by a block in which the development of maximal strength is the main aim.

A previous meta-analysis by Williams et al. [12] found that periodized resistance training was accompanied by larger increases in maximal strength when compared to non-periodized (NP) resistance training. However, the meta-analysis by Williams et al. did not equate for training volume between training groups. Thus, the inclusion of single-set NP training groups compared to multiple-set periodized training groups may have affected the result, since multiple-set protocols are superior to single-set protocols for development of maximal strength [7, 8, 13]. Furthermore, although previous studies have compared multiple periodization models, the findings are mixed—in some instances even contradictory. Hence, it remains unclear whether specific periodization models have differential effects on training-induced changes in muscular strength and muscle hypertrophy, and whether training status affects the potential effects of periodization. Therefore, investigating if there is an effect of periodization when volume is equated, and how intensity and volume are more effectively manipulated over time in order to augment gains in muscular strength and muscle hypertrophy, in both untrained and trained individuals, could potentially aid in maximizing strength development in both applied and clinical settings.

Consequently, the aims of this review and meta-analysis were to (a) examine the current body of literature to determine the effects of periodization of volume and intensity on strength and muscle hypertrophy, when volume is equated between conditions, (b) to determine if the effects of periodization are affected by training status, and (c) to determine how volume and intensity are more effectively periodized with the aim of maximizing strength gains and muscle hypertrophy, when the volume of resistance training is equated.

2 Methods

2.1 Search Strategy

The review was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [14]. Studies published before 7 February 2021 were located using searches of PubMed, Scopus, and SPORTDiscus databases. Combinations of the following search terms were used: "periodization," "periodized," "non-periodized," "periodization," "periodized," "non-periodised," "resistance training," "strength training," "weightlifting," "powerlifting," "strength," "muscular strength," "maximal strength," "1RM," "muscle hypertrophy," "muscle mass," "muscle cross-sectional area," "lean body mass," "fat-free mass." Duplicate publications were removed, and reference lists from retrieved articles were manually reviewed for additional publications not discovered based on database searches.

2.2 Study Selection

The inclusion criteria were as follows: Peer-reviewed publication; available in English; a comparison of at least one periodized resistance training group to either a NP resistance training group or another periodized resistance training group, using a different periodization model; training protocols involved dynamic resistance exercises with both concentric and eccentric muscle actions; loading of resistance exercises was determined relative to participants' level of strength; maximal strength measured by 1RM and/or at least one method of assessing changes in muscle mass; human participants; reporting of mean, standard deviation (SD), or standard error of the mean (SEM), and sample number provided in the text, table(s), or figure(s); study duration stated in weeks or months; equated for total volume (sets × repetitions) and mean intensity between at least two training groups; training protocols involved identical training frequency, exercises, exercise order, and execution for the different groups; training intervention with a duration of minimum 4 weeks.

A total of 1099 articles were identified from the initial search. Four other articles were identified through other sources. To reduce potential selection bias, the 1103 studies were independently reviewed by three of the investigators (LM, MB, and LC). A mutual decision was made about whether each study met the inclusion criteria. Any disagreements between reviewers were settled by consensus and consultation with a fourth investigator (JLJ). For articles that met the inclusion criteria but where data required to calculate effect sizes (ES) were not available, the corresponding authors were contacted by email and requested to provide the data. Where data were not presented in tables or text, and no additional data were provided by the authors, data were extracted from figures using WebPlotDigitizer (WebPlotDigitizer, V.4.4. Texas, USA: Ankit Rohatgi, 2020). Studies in which data extraction was not possible were excluded.

Studies were excluded for the following reasons: training groups used same periodization model, where the influence of other variables were investigated; only one training group was included in the study; did not use dynamic resistance exercises; no measures of muscle hypertrophy or 1RM; did not equate for volume and mean intensity between training groups; training groups did not use identical training frequency, exercises, or exercise order; loading not relative to participants' level of strength; used non-original data from already included studies; data to calculate ES not available. Figure 1 depicts a flow chart of the study selection process.

2.3 Coding of Studies

All of the included studies were individually coded by three of the investigators (LM, MB, LC) for the following variables: study characteristics (author(s), title and year of publication); participant characteristics, including age (classified as adolescents (< 18 years), adults (18–49 years), or older adults (≥ 50 years)), sex, resistance training status (as classified by the author(s) of the individual studies); description of training groups (classified as NP if there was no manipulation of volume and intensity throughout the training intervention; classified as LP if there was a linear increase in intensity and linear decrease in volume throughout the training intervention; classified as RLP if there was a linear decrease in intensity and linear increase in volume throughout the training intervention; classified as BP if the training intervention was divided into blocks with differences in intensity and volume, and intensity and volume did not linearly increase and decrease, respectively, throughout the training intervention; classified as WUP if there were non-linear weekly undulations in volume and intensity throughout the training intervention and classified as DUP if there were daily undulations in volume and intensity throughout the training intervention); study duration; the number of participants in each group; training variables, including training volume (number of sets per exercise and number of repetitions per set), training intensity and training frequency; 1RM tests; methods used for assessment of muscle hypertrophy.

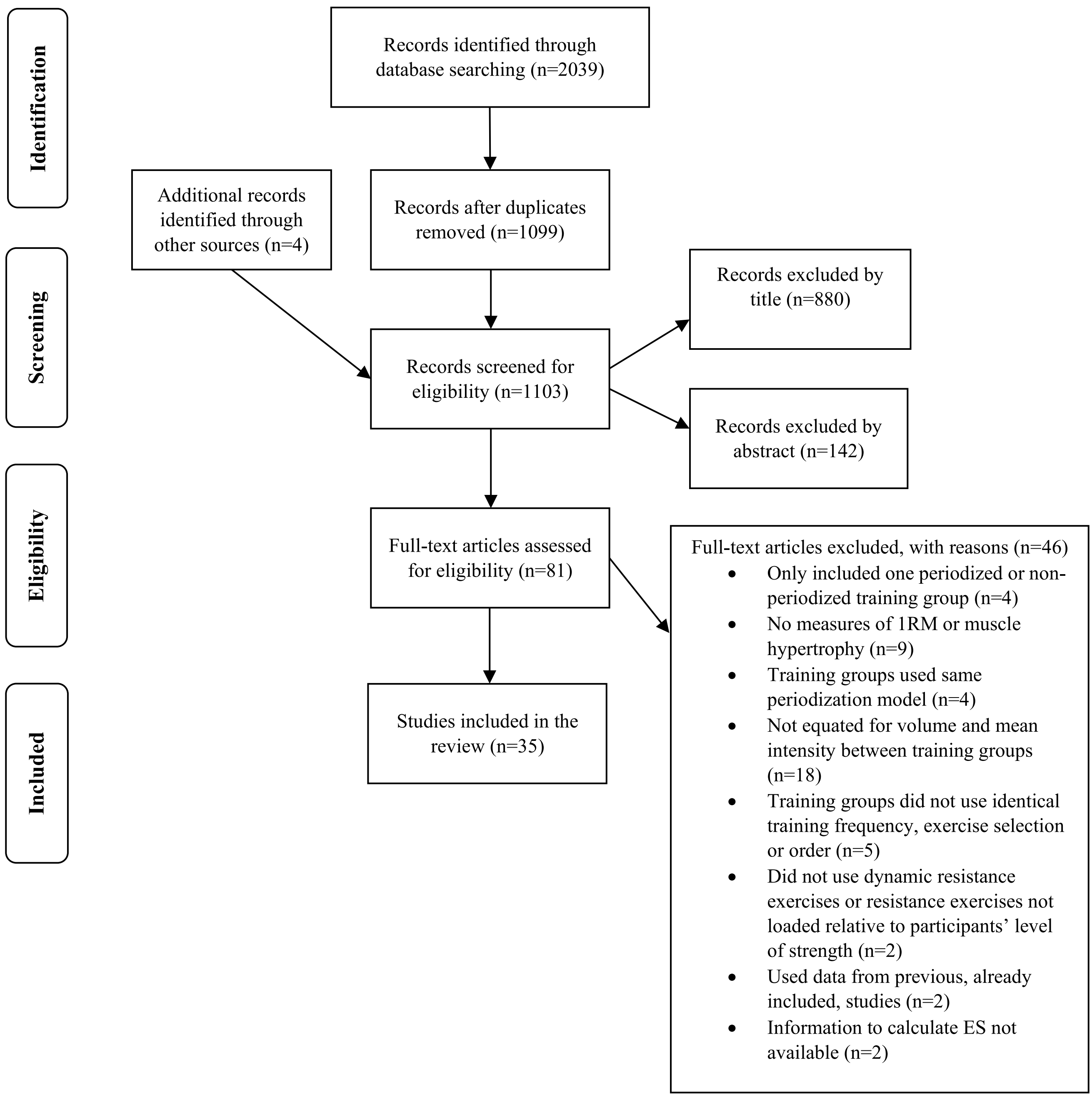

Fig. 1 Flow chart of study selection. *n* Number of studies, *1RM* 1-repetetion maximum, *ES* effect size

Coding was cross-checked between coders, and any discrepancies were resolved by mutual consensus and consultation with a fourth investigator (JLJ), if needed.

2.4 Bias Assessment

An assessment of the methodological quality of the included studies was performed using a modified PEDro scale. Details about the PEDro scale are presented elsewhere [15]. Because blinding of participants and investigators in exercise interventions is not feasible, no included studies could earn points on the PEDro scale items 5 and 6. Therefore, the scale was modified, so the maximum score was 8 points. The assessments of methodological quality were performed individually by three of the investigators (LM, MB, and LC), and any discrepancies were resolved by mutual consensus and consultation with a fourth investigator (JLJ), if needed.

2.5 Calculation of Effect Size

For each 1RM and muscle hypertrophy outcome an ES was calculated. Within-group ESs were calculated as the pretest–post-test change, divided by the pooled pretest SD [16]. When studies presented data as mean ± SEM, SEM was converted to SD. Between-group ESs were calculated as follows: $d = (\Delta M\text{t} - \Delta M\text{c})/\text{SDpre}$, where d is the magnitude of the ES, ΔMt is the pretest–posttest change of the treatment group, ΔMc is the pretest–posttest change of the control group and SDpre is the pooled pretest SD [16]. The variance around each ES was calculated using the sample size in each study and mean ES across all studies in the respective meta-analysis [17]. When studies assessed multiple 1RM outcomes between-group ESs were combined, accounting for non-independent ESs within studies as follows:

$V_Y = \frac{1}{m}V(1 + (m-1) \times r)$, where V_Y is the variance of the mean ES, V is the variance of the 1RM outcome ESs, m is the number of 1RM outcomes within the study and r is the correlation coefficient that describes the extent to which the outcomes correlate [18]. The correlation coefficients were calculated from the raw data of the included study by Eifler [19], which consisted of raw data from 200 participants and eight 1RM outcomes. For the few studies that assessed multiple muscle hypertrophy outcomes, raw data of the individual participants were not available. Therefore, the mean ES and corresponding variance was used.

Mean percent changes in 1RMs and measures of muscle mass in the individual studies were also calculated. Additionally, to control for differences between studies in the duration of the intervention (with longer training interventions being expected to result in larger increases in strength and muscle mass), relative changes per week were calculated, by dividing mean percent changes with the study duration.

2.6 Statistical Analysis

Four meta-analyses were performed: (a) comparison of changes in 1RM, in response to NP versus periodized resistance training; (b) comparison of measure(s) of muscle hypertrophy in response to NP versus periodized resistance training; (c) comparison of changes in 1RM, in response to LP versus UP; (d) comparison of measure(s) of muscle hypertrophy in response to LP versus UP. In the meta-analyses investigating LP versus UP, training groups coded as WUP and DUP were classified as UP. No meta-analyses of other periodization model comparisons were performed due to a low number of studies comparing these periodization models. All meta-analyses were performed using the RevMan 5.4 software (Review Manager (RevMan) Version 5.4. Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration, 2020). Between-group ESs with 95% confidence intervals (95% CIs) were used to determine effect measures, with an ES of 0.2, 0.5, and 0.8 representing small, moderate, and large differences, respectively [20]. The inverse-variance random effects model was used for the meta-analysis procedures, as the included studies were performed in different populations using diverse methods. Statistical heterogeneity of the included studies was assessed using the χ^2 and I^2 statistics, in which values of < 25% indicate low risk of heterogeneity, 25–75% indicate moderate risk of heterogeneity, and > 75% indicate high risk of heterogeneity [21]. To identify the presence of influential studies that might bias the analysis, sensitivity analyses were conducted by removing one study at a time and then examining the outcomes. Studies were identified as influential if removal resulted in change in P value from $P \leq 0.01$ to $P \geq 0.05$ or from $P \leq 0.05$ to $P \geq 0.10$, or vice versa. A subgroup analysis was performed to investigate if the results were affected by the training status of the participants. For this analysis, studies conducted on untrained and trained participants were analyzed separately. Subgroup analyses investigating potential effects of age and sex were not performed due to a low number of studies with only female, older adults or adolescent participants, in the respective meta-analyses. The statistical significance threshold was set at $P \leq 0.05$. Data are reported as mean, SD, and 95% CI.

3 Results

A total of 35 studies were included in the review, with a total of 1187 participants. 169 participants were classified as adolescents, 893 participants were classified as adults and 125 participants were classified as older adults. Thirteen studies included non-resistance training control groups that were not included in the analysis. Also, the study by Stone et al. [22] included a periodized resistance training group that was not included in the analysis, because volume was not equated with the other training groups. Thus, data from 1022 participants (129 classified as adolescents, 791 classified as adults, and 102 classified as older adults) were included in the calculation of ESs and percent changes. Nineteen studies included only male participants, seven studies included only female participants, and nine studies included participants of both sexes. Nineteen studies were conducted on untrained participants, 15 studies were conducted on trained participants, and one study was conducted on detrained participants. Study duration ranged from 6 to 36 weeks, with a mean study duration of 13.7 ± 6.0 weeks. Training frequency ranged from 2 to 4 days per week, with a mean training frequency of 2.9 ± 0.7 days per week. Prescribed volume per exercise ranged from 1 to 7 sets of 1–30 repetitions. Training intensity varied between 30–105%1RM and 3–30RM. Fifteen studies reported compliance to the training intervention, mean compliance to the training interventions were 93.0 ± 10.4% in these studies. Maximal strength was assessed by 1RM in 32 studies, using a variety of different resistance exercises, resulting in 210 within-group ESs. Muscle hypertrophy was assessed in 25 studies using a variety of methods, resulting in 64 within-group ESs. A summary of the included studies is presented in Table 1.

3.1 Methodological Quality of Studies

The values obtained for assessment of methodological quality using the PEDro scale are presented in Table 1. Across studies, PEDro scores ranged from 2 to 7 points with a mean PEDro score of 4.8 ± 1.0 points. A PEDro score of ≤ 3 points was interpreted as low methodological quality, a PEDro score of 4–5 points was interpreted as moderate methodological quality and a PEDro score of ≥ 6 points was interpreted as high methodological quality. Based on this interpretation, two studies were deemed to be of low methodological quality, 24 studies were deemed to be of moderate methodological quality, and nine studies were deemed to be of high methodological quality.

3.2 Non-periodized Versus Periodized Resistance Training

Fifteen studies compared NP to periodized resistance training, using various periodization models. Ten studies were conducted on untrained participants and five studies were conducted on trained participants.

3.2.1 Effects on Maximal Strength

All studies comparing NP to periodized resistance training assessed increases in maximal strength by 1RM. The mean relative change in 1RM across studies comparing NP to periodized resistance training was 22.7% ± 14.6% corresponding to 1.77% ± 1.06% per week for NP resistance training and 27.6% ± 18.5% corresponding to 2.13% ± 1.32% per week for periodized resistance training. The mean within-group ES for changes in 1RM across studies comparing NP to periodized resistance training was 0.98 ± 0.70 for NP resistance training and 1.30 ± 1.11 for periodized resistance training.

The meta-analysis comparing changes in 1RM between NP and periodized training groups comprised of 70 between-group ESs from the 15 studies (Fig. 2). The studies included in the meta-analysis were found to be moderately heterogeneous ($\chi^2 = 51.47$, $df = 14$ ($P < 0.00001$), $I^2 = 73\%$).

The meta-analysis indicated a significant effect on 1RM favoring periodized training (ES 0.31, 95% CI [0.04, 0.57]; Z = 2.28 ($P = 0.02$)). When removing one study at a time, no sensitivity analyses revealed influential studies that altered the *P* value. However, in the sensitivity analysis in which Monteiro et al. [42] was removed the magnitude of the effect decreased (ES 0.19, 95% CI [0.01, 0.38]; Z = 2.05 (P = 0.04)), and observed heterogeneity was reduced to a degree where it was not significant ($\chi^2 = 20.89$, $df = 13$ ($P = 0.08$), $I^2 = 38\%$). No significant differences were found when studies conducted on untrained or trained participants were analyzed separately.

3.2.2 Effects on Muscle Hypertrophy

Ten studies comparing NP to periodized resistance training assessed muscle hypertrophy. Various methods were used for assessment of muscle hypertrophy (Table 1). The mean relative change in measures of muscle mass across studies comparing NP to periodized resistance training was 3.2% ± 3.2% corresponding to 0.27% ± 0.31% per week for NP resistance training and 4.2% ± 3.4% corresponding to 0.34% ± 0.34% per week for periodized resistance training. The mean within-group ES for changes in measures of muscle mass across studies comparing NP to periodized resistance training was 0.22 ± 0.30 for NP resistance training and 0.34 ± 0.28 for periodized resistance training.

The meta-analysis comparing effects on measures of muscle hypertrophy between NP and periodized resistance training comprised of 16 between-group ESs from the ten studies (Fig. 3). The studies included in the meta-analysis were found not to be heterogenous ($\chi^2 = 3.73$, $df = 9$ ($P = 0.93$), $I^2 = 0\%$). The meta-analysis showed no clear effect on muscle hypertrophy favoring NP or periodized resistance training (ES 0.13, 95% CI [–0.10, 0.36]; $Z = 1.10$ ($P = 0.27$)). None of the sensitivity analyses removing one study at a time revealed influential studies that altered the results. In addition, no significant differences were found when studies conducted on untrained or trained participants were analyzed separately.

3.3 Linear Periodization (LP) versus Undulating Periodization (UP)

Sixteen studies compared LP to UP, periodized as DUP, WUP, or a combination of the two. Nine studies were conducted on untrained participants, six studies were conducted on trained participants, and one study was conducted on detrained participants.

3.3.1 Effects on Maximal Strength

Fourteen of the studies comparing LP to UP assessed maximal strength by 1RM. The mean relative change in 1RM across studies comparing LP to UP was 17.8% ± 11.2% corresponding to 1.71% ± 1.29% per week for LP and 22.8% ± 12.5% corresponding to 2.20% ± 1.43% per week for UP. The mean within-group ES for changes in 1RM

Table 1 Overview of included studies

Study	Participants	Comparison(s)	Sets × reps; intensity	Study duration; training frequency	Measures of 1RM and muscle hypertrophy	Mean 1RM percent change; mean 1RM ES	Mean muscle mass percent change; mean muscle hypertrophy ES	PEDro score
Ahmadizad et al. [23]	32 untrained adult males	NP ($n=8$) vs LP ($n=8$) vs. DUP ($n=8$)	NP: 2 × 12; 70%1RM LP: 2 × 8–18; 50–85%1RM DUP: 2 × 8–16; 55–85%1RM	8 weeks; 3 days/week	1RM bench press 1RM squat BIA	NP: 6.0% (0.75%/week); 0.34 LP: 10.5% (1.31%/week); 0.63 DUP: 12.4% (1.55%/week); 0.74	NP: 1.9% (0.24%/week); 0.19 LP: 1.9% (0.24%/week); 0.19 DUP: 2.0% (0.25%/week); 0.19	5
Baker et al. [24]	22 untrained adult males	NP ($n=9$) vs LP ($n=8$) vs. WUP ($n=5$)	NP: 3–5 × 6–8; 6–8RM LP: 3–5 × 3–10; 3–10RMs WUP: 3–5 × 3–10; 3–10RMs	12 weeks; 3 days/week	1RM squat 1RM bench press skinfolds	NP: 19.3% (1.61%/week); 0.98 LP: 19.6% (1.63%/week); 0.94 WUP: 22.4% (1.87%/week); 1.04	NP: 3.2% (0.27%/week); 0.34 LP: 2.8% (0.23%/week); 0.28 WUP: 3.4% (0.28%/week); 0.31	3
Bartolomei et al. [25]	24 trained adult males	BP ($n=14$) vs. WUP ($n=10$)	BP: NR × 1–10; 50–95%1RM WUP: 5 × 3–10; 50–95%1RM	15 weeks; 4 days/week	1RM bench press skinfolds	BP: 7.6% (0.51%/week); 0.37 WUP: 2.0% (0.13%/week); 0.09	BP: 3.2% (0.21%/week); 0.36 WUP: 2.0% (0.13%/week); 0.22	5
Bartolomei et al. [26]	17 trained adult females	BP ($n=9$) vs. WUP ($n=8$)	BP: 3–5 × 3–10; 65–93%1RM WUP: 3–5 × 3–10; 65–93%1RM	10 weeks; 3 days/week	1RM squat 1RM bench press 1RM deadlift skinfolds circumference measures	BP: 11.6% (1.16%/week); 0.56 WUP: 17.3% (1.73%/week); 0.78	BP: 5.5% (0.55%/week); 0.36 WUP: 6.1% (0.61%/week); 0.43	5
Bartolomei et al. [27]	18 trained adult males	BP ($n=10$) vs. WUP ($n=8$)	BP: NR × 1–10; 50–95%1RM WUP: 4–5 × 3–10; 50–95%1RM	15 weeks; 4 days/week	1RM bench press skinfolds	BP: 7.5% (0.50%/week); 0.44 WUP: 1.3% (0.09%/week); 0.07	BP: 2.5% (0.17%/week); 0.22 WUP: 1.7% (0.11%/week); 0.15	5
Buford et al. [28]	28 detrained adult males and females	LP ($n=9$) vs. DUP ($n=10$) vs. WUP ($n=9$)	LP: 3 × 4–8; 80–90%1RM DUP: 3 × 4–8; 80–90%1RM WUP: 3 × 4–8; 80–90%1RM	9 weeks; 3 days/week	1RM bench press 1RM leg press	LP: 54.8% (6.09%/week); 1.61 DUP: 48.3% (5.37%/week); 1.57 WUP: 62.1% (6.90%/week); 1.81	Not assessed	5

Table 1 (continued)

Study	Participants	Comparison(s)	Sets × reps; intensity	Study duration; training frequency	Measures of 1RM and muscle hypertrophy	Mean 1RM percent change; mean 1RM ES	Mean muscle mass percent change; mean muscle hypertrophy ES	PEDro score
Conlon et al. [29]	33 untrained older adult males and females	NP ($n=10$) vs. BP ($n=13$) vs. DUP ($n=10$)	NP: 3 × 10; 10RM BP: 3 × 5–15; 5–15RMs DUP: 3 × 5–15; 5–15RMs	22 weeks; 3 days/week	1RM leg press 1RM chest press DEXA	NP: 33.7% (1.53%/week); 0.91 BP: 46.8% (2.13%/week); 1.19 DUP: 32.8% (1.49%/week); 1.09	NP: 7.2% (0.33%/week); 0.32 BP: 5.6% (0.25%/week); 0.24 DUP: 7.6% (0.35%/week); 0.36	5
DeBeliso et al. [30]	43 untrained older adult males and females	NP ($n=13$) vs. LP ($n=17$)	NP: 3 × 9; 9RM LP: 2–4 × 6–15; 6–15RM	18 weeks; 2 days/week	1RM leg press 1RM leg extension 1RM leg curl 1RM chest press 1RM shoulder press 1RM lat pulldown 1RM triceps extension 1RM biceps curl	NP: 51.2% (2.84%/week); 1.33 LP: 61.6% (3.42%/week); 1.39	Not assessed	4
De Lima et al. [31]	28 untrained adult females	DUP ($n=10$) vs. WUP ($n=10$)	DUP: 3 × 15–30; 15–30RM WUP: 3 × 15–30; 15–30RM	12 weeks; 4 days/week	1RM bench press 1RM leg press 1RM biceps curl skinfolds	DUP: 28.1% (2.34%/week); 1.52 WUP: 29.4% (2.45%/week); 1.53	DUP: 3.5% (0.29%/week); 0.32 WUP: 4.7% (0.39%/week); 0.43	4
De Souza et al. [32]	33 untrained adult males	NP ($n=8$) vs. LP ($n=9$) vs. DUP ($n=8$)	NP: 2–3 × 8; 8RM LP: 2–4 × 4–12; 4–12RM DUP: 2–4 × 4–12; 4–12RM	12 weeks; 2 days/week	1RM squat MRI	NP: 21.4% (1.78%/week); 1.14 LP: 17.9% (1.49%/week); 0.95 DUP: 19.2% (1.60%/week); 1.09	NP: 9.2% (0.77%/week); 0.74 LP: 11.4% (0.95%/week); 0.91 DUP: 11.8% (0.98%/week); 0.91	5
Eifler [19]	200 trained adult males and females	NP ($n=50$) vs. LP ($n=50$) vs. RLP ($n=50$) vs. DUP ($n=50$)	NP: 3 × 10; 80%1RM LP: 3 × 5–15; 70–90%1RM RLP: 3 × 5–15; 70–90%1RM DUP: 3 × 5–15; 70–90%1RM	6 weeks; 3 days/week	1RM leg press 1RM chest press 1RM fly 1RM lat pulldown 1RM row 1RM shoulder press 1RM triceps push-down 1RM biceps curl	NP: 13.0% (2.17%/week); 0.32 LP: 14.9% (2.48%/week); 0.35 RLP: 14.2% (2.37%/week); 0.37 DUP: 23.6% (3.93%/week); 0.53	Not assessed	4

Table 1 (continued)

Study	Participants	Comparison(s)	Sets × reps; intensity	Study duration; training frequency	Measures of 1RM and muscle hypertrophy	Mean 1RM percent change; mean 1RM ES	Mean muscle mass percent change; mean muscle hypertrophy ES	PEDro score
Foschini et al. [33]	32 untrained adolescent males and females	LP ($n=16$) vs. DUP ($n=16$)	LP: 3 × 6–20; 6–20RMs DUP: 3 × 6–20; 6–20RMs	14 weeks; 3 days/week	BOD POD	Not assessed	LP: 3.4% (0.24%/week); 0.22 DUP: 2.4% (0.17%/week); 0.17	6
Franchini et al. [34]	13 trained adult males	RLP ($n=6$) vs. DUP ($n=7$)	RLP: 4 × 3–20; 3–20RMs DUP: 4 × 3–20; 3–20RMs	8 weeks; 3 days/week	1RM squat 1RM bench press 1RM row	RLP: 10.0% (1.25%/week); 0.62 DUP: 9.4% (1.18%/week); 0.56	Not assessed	4
Gavanda et al. [35]	28 trained adolescent males	LP ($n=14$) vs. DUP ($n=14$)	LP: 1–4 × 5–20; 55–85%1RM DUP: 1–4 × 5–20; 55–85%1RM	12 weeks; 3 days/week	1RM squat 1RM bench press BIA ultrasound	LP: 16.2% (1.35%/week); 0.60 DUP: 15.9% (1.33%/week); 0.55	LP: 7.0% (0.58%/week); 0.43 DUP: 7.3% (0.61%/week); 0.43	4
Harries et al. [36]	26 trained adolescent males	DUP ($n=8$) vs. WUP ($n=8$)	DUP: 4–6 × 3–10; 60–81%1RM WUP: 4–6 × 3–10; 60–81%1RM	12 weeks; 2 days/week	BIA	Not assessed	DUP: 1.0% (0.08%/week); 0.06 WUP: 2.1% (0.18%/week); 0.13	7
Herrick and Stone [37]	20 untrained adult females	NP ($n=10$) vs. LP ($n=10$)	NP: 3 × 6; 6RM LP: 3 × 3–10; 3–10RMs	15 weeks; 2 days/week	1RM squat 1RM bench press	NP: 35.8% (2.39%/week); 1.28 LP: 42.7% (2.85%/week); 1.45	Not assessed	5
Inoue et al. [38]	45 untrained adolescent males and females	LP ($n=13$) vs. DUP ($n=12$)	LP: 3 × 6–20; 6–20RMs DUP: 3 × 6–20; 6–20RMs	26 weeks; 3 days/week	BOD POD	Not assessed	LP: 7.9% (0.30%/week); 0.60 DUP: 2.6% (0.10%/week); 0.22	4
Kok et al. [39]	20 untrained adult females	BP ($n=10$) vs. DUP ($n=10$)	BP: 3–4 × 6–10; 30–90%1RM DUP: 3–4 × 6–10; 30–90%1RM	9 weeks; 3 days/week	1RM squat 1RM bench press ultrasound	BP: 28.3% (3.14%/week); 1.36 DUP: 34.8% (3.87%/week); 1.55	BP: 8.7% (0.97%/week); 0.57 DUP: 14.8% (1.64%/week); 0.87	5
Kraemer et al. [40]	27 untrained adult females	NP ($n=10$) vs. DUP ($n=9$)	NP: 2–3 × 8–10; 8–10RM DUP: 2–3 × 4–15; 4–15RMs	36 weeks; 3 days/week	1RM bench press 1RM leg press 1RM shoulder press skinfolds	NP: 20.2% (0.56%/week); 0.97 DUP: 23.9% (0.66%/week); 1.23	NP: 3.5% (0.10%/week); 0.36 DUP: 7.1% (0.20%/week); 0.74	5
Miranda et al. [41]	20 trained adult males	LP ($n=10$) vs. DUP ($n=10$)	LP: 3 × 4–10; 4–10RM DUP: 3 × 4–10; 4–10RM	12 weeks; 4 days/week	1RM bench press 1RM leg press	LP: 12.5% (1.04%/week); 0.60 DUP: 17.0% (1.42%/week); 1.66	Not assessed	6

Table 1 (continued)

Study	Participants	Comparison(s)	Sets × reps; intensity	Study duration; training frequency	Measures of 1RM and muscle hypertrophy	Mean 1RM percent change; mean 1RM ES	Mean muscle mass percent change; mean muscle hypertrophy ES	PEDro score
Monteiro et al. [42]	27 trained adult males	NP ($n=9$) vs. LP ($n=9$) vs. DUP ($n=9$)	NP: 3 × 8–10; 8–10RM LP: 3 × 4–15; 4–15RMs DUP: 3 × 4–15; 4–15RMs	12 weeks; 4 days/week	1RM bench press 1RM leg press skinfolds	NP: 8.5% (0.71%/week); 0.86 LP: 10.4% (0.87%/week); 1.13 DUP: 35.6% (2.97%/week); 3.83	NP: –3.0% (–0.25%/week); –0.50 LP: 1.2% (0.10%/week); 0.19 DUP: 0.3% (0.03%/week); 0.05	4
Moraes et al. [43]	38 untrained adolescent males	NP ($n=14$) vs. DUP ($n=14$)	NP: 3 × 10–12; 10–12RM DUP: 3 × 3–20; 3–20RMs	12 weeks; 3 days/week	1RM bench press 1RM leg press	NP: 53.6% (4.47%/week); 3.30 DUP: 71.2% (5.93%/week); 4.62	Not assessed	6
Peterson et al. [44]	14 trained adult males	BP ($n=7$) vs. DUP ($n=7$)	BP: 2–7 × NR; 30–100%1RM DUP: 2–7 × NR; 30–100%1RM	9 weeks; 3 days/week	1RM squat 1RM bench press	BP: 14.4% (1.60%/week); 0.57 DUP: 18.7% (2.08%/week); 0.90	Not assessed	6
Prestes et al. [45]	40 trained adult males	DUP ($n=20$) vs. WUP ($n=20$)	DUP: 3 × 6–12; 6–12RMs WUP: 3 × 6–12; 6–12RMs	12 weeks; 4 days/week	1RM bench press 1RM leg press 1RM biceps curl	DUP: 29.7% (2.48%/week); 1.19 WUP: 19.0% (1.58%/week); 0.80	Not assessed	6
Prestes et al. [46]	49 untrained older adult females	LP ($n=20$) vs. DUP ($n=19$)	LP: 3 × 6–14; 6–14RMs DUP: 3 × 6–14; 6–14RMs	16 weeks; 2 days/week	1RM bench press 1RM leg press 1RM biceps curl DEXA	LP: 25.0% (1.56%/week); 0.86 DUP: 13.6% (0.85%/week); 0.47	LP: –1.6% (–0.10%/week); –0.14 DUP: 1.7% (0.11%/week); 0.15	4
Rhea et al. [47]	20 trained adult males	LP ($n=10$) vs. DUP ($n=10$)	LP: 3 × 4–8; 4–8RMs DUP: 3 × 4–8; 4–8RMs	12 weeks; 3 days/week	1RM bench press 1RM leg press	LP: 20.0% (1.67%/week); 0.87 DUP: 42.3% (3.53%/week); 1.51	Not assessed	6
Rhea et al. [48]	60 trained adult males and females	LP ($n=20$) vs. RLP ($n=20$) vs. DUP ($n=20$)	LP: 3 × 15–25; 15–25RMs RLP: 3 × 15–25; 15–25RMs DUP: 3 × 15–25; 15–25RMs	15 weeks; 2 days/week	1RM leg extension	LP: 9.1% (0.61%/week); 0.31 RLP: 5.6% (0.37%/week); 0.17 DUP: 9.8% (0.65%/week); 0.31	Not assessed	5

Table 1 (continued)

Study	Participants	Comparison(s)	Sets × reps; intensity	Study duration; training frequency	Measures of 1RM and muscle hypertrophy	Mean 1RM percent change; mean 1RM ES	Mean muscle mass percent change; mean muscle hypertrophy ES	PEDro score
Schiotz et al. [49]	14 trained adult males	NP ($n=8$) vs. LP ($n=6$)	NP: 4 × 6; 80%1RM LP: 5–6 × 1–10; 50–105%1RM	10 weeks; 4 days/week	1RM squat 1RM bench press skinfolds	NP: 8.1% (0.81%/week); 0.37 LP: 9.0% (0.90%/week); 0.40	NP: 0.4% (0.04%/week); 0.03 LP: 1.1% (0.11%/week): 0.08	4
Simão et al. [50]	30 untrained adult males	LP ($n=10$) vs. DUP/WUP ($n=11$)	LP: 2–4 × 3–15; 3–15RMs DUP/WUP: 2–4 × 3–15; 3–15RMs	12 weeks; 2 days/week	1RM bench press 1RM lat pulldown 1RM biceps curl 1RM triceps extension ultrasound	LP: 14.9% (1.24%/week); 0.83 DUP/WUP: 18.6% (1.55%/week); 1.08	LP: 3.3% (0.28%/week); 0.21 DUP/WUP: 6.7% (0.56%/week); 0.48	6
Soares et al. [51]	41 untrained adult males and females	NP ($n=13$) vs. DUP ($n=13$)	NP: 3 × 8–12; 70%1RM DUP: 3 × 4–20; 50–90%1RM	12 weeks; 3 days/week	1RM bench press 1RM leg press 1RM lat pulldown 1RM leg curl 1RM triceps push-down 1RM calf raise BIA	NP: 29.3% (2.44%/week); 0.59 DUP: 37.9% (3.16%/week); 1.03	NP: 1.9% (0.16%/week); 0.16 DUP: 2.5% (0.21%/week); 0.22	5
Souza et al. [52]	31 untrained adult males	NP ($n=9$) vs. LP ($n=9$) vs. DUP ($n=8$)	NP: 2–3 × 8; 8RM LP: 2–4 × 8–12; 8–12RMs DUP: 2–4 × 6–12; 6–12RMs	6 weeks; 2 days/week	1RM squat MRI	NP: 17.1% (2.85%/week); 0.91 LP: 8.4% (1.40%/week); 0.45 DUP: 11.9% (1.98%/week); 0.67	NP: 5.1% (0.85%/week); 0.41 LP: 4.7% (0.78%/week); 0.38 DUP: 7.8% (1.30%/week); 0.60	5
Spineti et al. [53]	32 untrained adult males	LP ($n=13$) vs. DUP ($n=10$)	LP: 2–4 × 3–15; 3–15RMs DUP: 2–4 × 3–15; 3–15RMs	12 weeks; 2 days/week	1RM bench press 1RM lat pulldown 1RM biceps curl 1RM triceps extension ultrasound	LP: 14.9% (1.24%/week); 0.96 DUP: 21.7% (1.81%/week); 1.31	LP: 6.2% (0.52%/week); 0.48 DUP: 7.7% (0.64%/week); 0.65	4
Stone et al. [22]	21 trained adult males	NP ($n=5$) vs. LP ($n=9$)	NP: 5 × 6; 6RM LP: 3–4 × 3–10; 3–10RMs	12 weeks; 3 days/week	1RM squat	NP: 10.0% (0.83%/week); 0.76 LP: 14.9% (1.24%/week); 1.02	Not assessed	2
Streb et al. [54]	34 untrained adult males and females	NP ($n=8$) vs. LP ($n=11$)	NP: 2 ×10–12; 10–12RM LP: 2 × 8–14; 8–14RMs	16 weeks; 3 days/week	1RM bench press 1RM leg press BIA	NP: 13.4% (0.84%/week); 0.40 LP: 12.0% (0.75%/week); 0.35	NP: 2.7% (0.17%/week); 0.11 LP: 0.5% (0.03%/week); 0.02	6

Table 1 (continued)

Study	Participants	Comparison(s)	Sets × reps; intensity	Study duration; training frequency	Measures of 1RM and muscle hypertrophy	Mean 1RM percent change; mean 1RM ES	Mean muscle mass percent change; mean muscle hypertrophy ES	PEDro score
Vanni et al. [55]	27 untrained adult females	LP ($n=14$) vs. BP ($n=13$)	LP: 3 × 6–20; 6–20RMs BP: 3–4 × 6–20; 6–20RMs	28 weeks; 3 days/week	1RM bench press 1RM leg press 1RM hack squat 1RM hip abduction 1RM hip adduction 1RM abdominal chair 1RM waist cross over 1RM row	LP: 43.5% (1.55%/week); 2.08 BP: 44.9% (1.60%/week); 2.08	Not assessed	4

ES effect size, *NP* non-periodized, *LP* linear periodization, *RLP* reverse linear periodization, *BP* block periodization, *DUP* daily undulating periodization, *WUP* weekly undulating periodization, *RM* repetition maximum, *1RM* 1-repetetion maximum, *BIA* bioelectrical impedance analysis, *DEXA* dual-energy X-ray absorptiometry, *MRI* magnetic resonance imaging, *BOD POD* air displacement plethysmography, *NR* not reported.

across studies comparing LP to UP was 0.81 ± 0.35 for LP and 1.26 ± 1.12 for UP.

The meta-analysis comparing changes in 1RM between LP and UP comprised of 38 between-group ESs from the 14 studies (Fig. 4). The studies included in the meta-analysis were found to be moderately heterogeneous ($\chi^2 = 43.35$, df = 13 ($P < 0.00001$), $I^2 = 70\%$). The meta-analysis indicated a significant effect on 1RM favoring UP (ES 0.31, 95% CI [0.02, 0.61]; $Z = 2.06$ ($P = 0.04$)). When removing one study at a time, no sensitivity analyses revealed influential studies that altered the P value. However, in the sensitivity analysis in which Monteiro et al. [42] was removed the magnitude of the effect decreased twofold (ES = 0.15, 95% CI [0.00, 0.30]; $Z = 1.92$ ($P = 0.05$)), and observed heterogeneity was eliminated ($\chi^2 = 10.70$, $df = 12$ ($P = 0.56$), $I^2 = 0\%$). Subgroup analyses in which studies conducted on untrained and trained participants were analyzed separately indicated a moderate effect favoring UP for studies conducted on trained participants (ES 0.61, 95% CI [0.00, 1.22]; $Z = 1.97$ ($P = 0.05$)), whereas no effect favoring LP or UP was found for studies conducted on untrained participants (ES 0.06, 95% CI [–0.20, 0.31]; $Z = 0.43$ ($P = 0.67$)), indicating that training status may influence the effects of these periodization models.

3.3.2 Effects on Muscle Hypertrophy

Eleven studies comparing LP to UP assessed muscle hypertrophy. Various methods were used for assessment of muscle hypertrophy (Table 1). The mean relative change in measures of muscle mass across studies comparing LP to UP was 4.4% ± 3.4% corresponding to 0.37% ± 0.29% per week for LP and 4.9% ± 3.4% corresponding to 0.46% ± 0.39% per week for UP. The mean within-group ES for changes in measures of muscle mass across studies comparing LP to UP was 0.34 ± 0.26 for LP and 0.38 ± 0.25 for UP.

The meta-analysis comparing effects on measures of muscle hypertrophy between LP and UP comprised of 16 between-group ESs from the 11 studies (Fig. 5). The studies included in the meta-analysis were found not to be heterogeneous ($\chi^2 = 2.46$, $df = 10$ ($P = 0.99$), $I^2 = 0\%$). The meta-analysis showed no effect on muscle hypertrophy favoring LP or UP (ES 0.05, 95% CI [–0.20, 0.29]; $Z = 0.36$ ($P = 0.72$)). None of the sensitivity analyses removing one study at a time revealed influential studies that altered the results. In addition, no significant differences were found when studies conducted on untrained or trained participants were analyzed separately.

3.4 Other Periodization Model Comparisons

Six studies compared BP to UP, either periodized as DUP or WUP. All six studies assessed maximal strength by 1RM

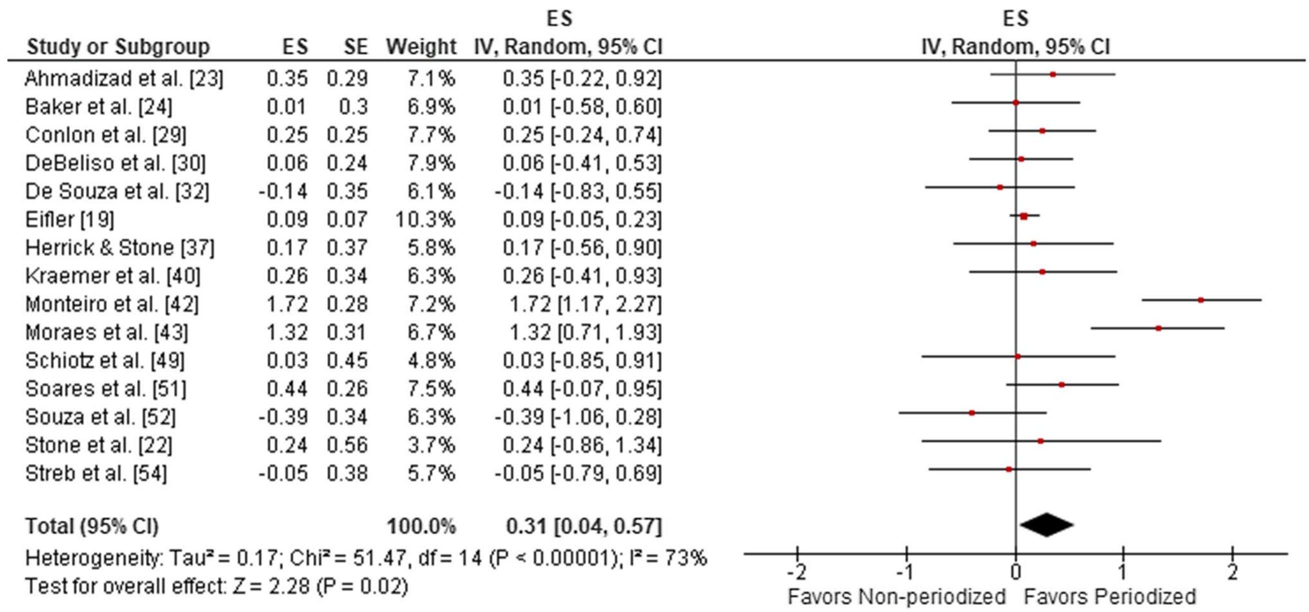

Fig. 2 Forest plot of changes in 1RM for non-periodized and periodized resistance training. *1RM* 1-repetition maximum, *ES* effect size, *SE* standard error, *IV* inverse variance, *CI* confidence interval

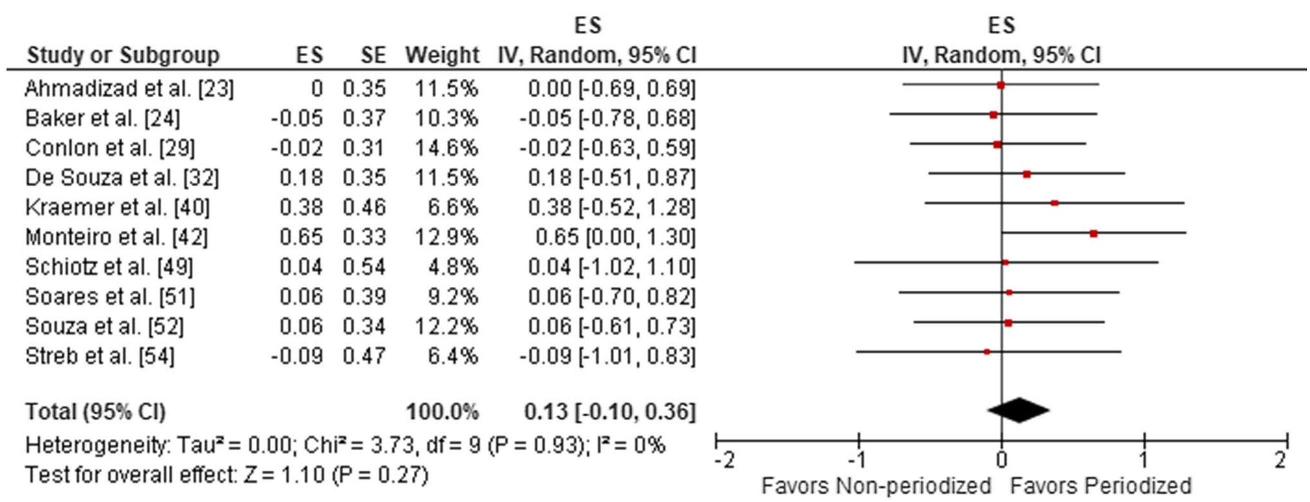

Fig. 3 Forest plot of measures of muscle hypertrophy for non-periodized and periodized resistance training. *ES* effect size, *SE* standard error, *IV* inverse variance, *CI* confidence interval

and five of the studies assessed muscle hypertrophy. Four studies compared DUP to WUP. Three of these studies assessed maximal strength by 1RM and two of the studies assessed muscle hypertrophy. Three studies compared DUP to RLP. All of the three studies assessed maximal strength by 1RM, whereas none of the studies assessed muscle hypertrophy. Two studies compared LP to RLP. Both of these studies assessed maximal strength by 1RM and did not assess muscle hypertrophy. The study by Vanni et al. [55] was the only study that compared LP to BP. This study assessed maximal strength by 1RM and did not assess muscle hypertrophy.

There was an insufficient number of studies investigating these periodization model comparisons to determine differences in effects on maximal strength and muscle hypertrophy based on meta-analyses. Results of the individual studies comparing these periodization models are presented in Table 1.

4 Discussion

The results from the present systematic review and meta-analysis show that when comparing the change from pre to post intervention between volume-equated resistance training programs, periodized resistance training leads to larger increases in maximal dynamic strength (1RM) as compared to NP resistance training. No significant differences were found when comparing measures of muscle hypertrophy between NP and periodized resistance training, indicating that muscle hypertrophy is similar between NP and periodized resistance training. Furthermore, UP was found to promote increases in strength more efficiently than LP, indicating that daily or weekly variations in volume and intensity are desirable when the aim is to maximize strength development. This effect was driven largely by trained as opposed to untrained individuals. In contrast, no significant differences were found when comparing measures of muscle hypertrophy between LP and UP, indicating that muscle hypertrophy in response to resistance training is not affected by differences in these periodization models.

The finding that periodized resistance training resulted in larger increases in 1RM than NP resistance training largely match the results of the meta-analysis by Williams et al. [12], in which an ES of 0.43 was found favoring periodized resistance training. In contrast to the results from Williams et al., the observed greater increases in maximal strength in response to periodized resistance training compared to NP resistance training cannot be explained by differences in training volume, as this is the first meta-analysis to only include studies in which volume has been equated between periodized and NP resistance training. Two meta-analyses have previously compared the effects of LP and UP on maximal strength. Harries et al. [56] found no significant differences between LP and UP on upper-body and lower-body strength, whereas Caldas et al. [57] found an ES of 0.22 favoring UP. Both of these meta-analyses computed standardized mean differences (SMD) between posttests, not accounting for within-group differences between pretest and posttest. This may explain discrepancies in results with the present meta-analysis. Although this is the first meta-analysis comparing muscle hypertrophy between NP and periodized resistance training, the findings that no differences in muscle hypertrophy were seen between NP and periodized resistance training largely match the conclusion of a recent systematic review by Grgic et al. [58]. A meta-analysis also by Grgic and co-workers [59] investigating muscle hypertrophy between LP and UP is also in agreement with the results of the present systematic review and meta-analysis, indicating that muscle hypertrophy is similar between LP and UP.

While the results of the present systematic review and meta-analysis demonstrate greater increases in maximal strength with periodized resistance training compared to NP resistance training and with UP compared to LP, the

Study or Subgroup	ES	SE	Weight	ES IV, Random, 95% CI
Ahmadizad et al. [23]	0.11	0.41	6.1%	0.11 [-0.69, 0.91]
Baker et al. [24]	0.1	0.47	5.4%	0.10 [-0.82, 1.02]
Buford et al. [28]	0.09	0.27	8.2%	0.09 [-0.44, 0.62]
De Souza et al. [32]	0.13	0.49	5.1%	0.13 [-0.83, 1.09]
Eifler [19]	0.19	0.13	10.3%	0.19 [-0.06, 0.44]
Gavanda et al. [35]	-0.06	0.31	7.6%	-0.06 [-0.67, 0.55]
Miranda et al. [41]	0.72	0.37	6.7%	0.72 [-0.01, 1.45]
Monteiro et al. [42]	2.42	0.39	6.4%	2.42 [1.66, 3.18]
Prestes et al. [46]	-0.39	0.24	8.7%	-0.39 [-0.86, 0.08]
Rhea et al. [47]	0.63	0.37	6.7%	0.63 [-0.10, 1.36]
Rhea et al. [48]	-0.01	0.32	7.5%	-0.01 [-0.64, 0.62]
Simão et al. [50]	0.25	0.29	7.9%	0.25 [-0.32, 0.82]
Souza et al. [52]	0.22	0.49	5.1%	0.22 [-0.74, 1.18]
Spineti et al. [53]	0.36	0.28	8.1%	0.36 [-0.19, 0.91]
Total (95% CI)			**100.0%**	**0.31 [0.02, 0.61]**

Heterogeneity: Tau² = 0.21; Chi² = 43.35, df = 13 (P < 0.0001); I² = 70%
Test for overall effect: Z = 2.06 (P = 0.04)

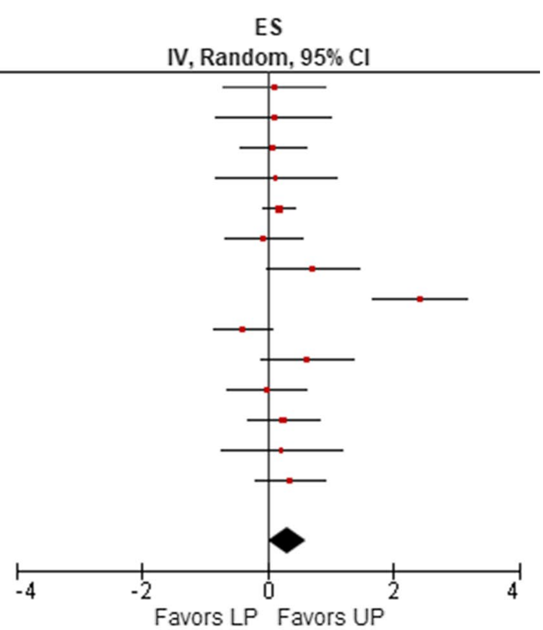

Fig. 4 Forest plot of changes in 1RM for linear periodization and undulating periodization. *1RM* 1-repetition maximum, *LP* linear periodization, *UP* undulating periodization, *ES* effect size, *SE* standard error, *IV* inverse variance, *CI* confidence interval

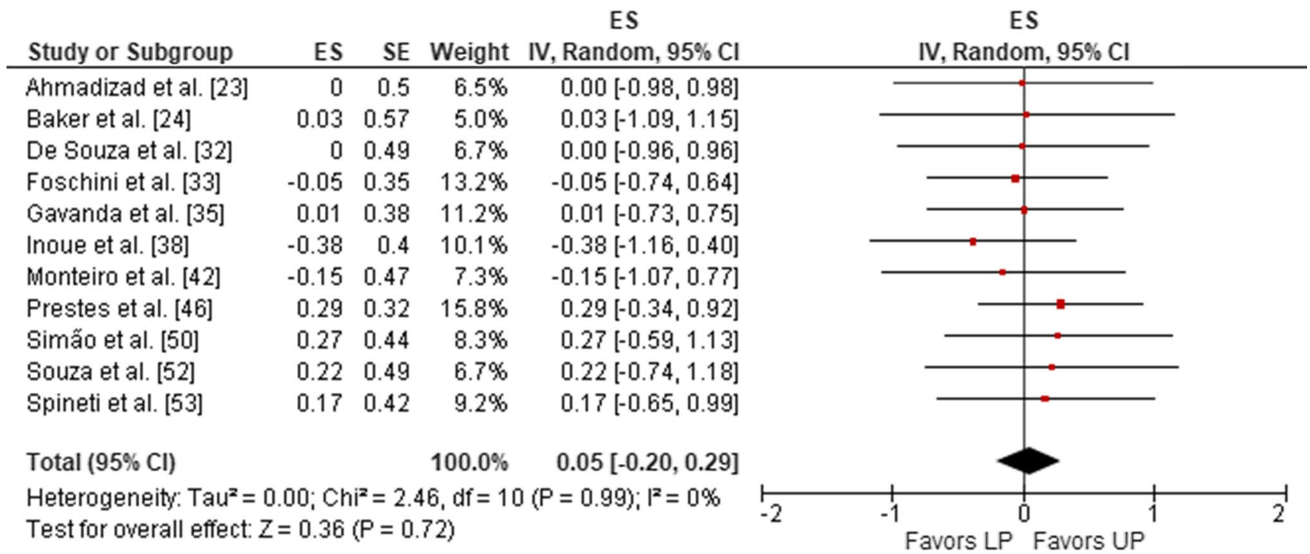

Fig. 5 Forest plot of measures of muscle hypertrophy for linear periodization and undulating periodization. *LP* linear periodization, *UP* undulating periodization, *ES* effect size, *SE* standard error, *IV* inverse variance, *CI* confidence interval

meta-analyses also indicate a moderate to high degree of heterogeneity between studies. Although caution should be taken in the case of heterogeneity, it should also be noted that in the sensitivity analyses in which Monteiro et al. [42] was removed heterogeneity was either reduced to non-significant levels or completely eliminated, indicating that this study may have biased the results. When heterogeneity was eliminated, small but significant effects favoring periodized resistance training and UP, respectively, remained present. In addition to investigating the effects of NP vs. periodized resistance training and LP vs. UP, this seems to be the first review to compare effects of other periodization models in regard to resistance training. Most of these studies investigating effects of other periodization models found no significant differences in changes in 1RM or measures of muscle hypertrophy between training groups. Unfortunately, additional future studies investigating effects of these periodization model comparisons are needed, in order to allow meta-analyses to be performed with sufficient statistical power.

4.1 Mechanisms Underlying the Effect of Periodization on Maximal Strength

There are a number of different factors that may contribute to increased strength, as a result of resistance training. These include not only morphological adaptations, such as muscle hypertrophy and increases in muscle fiber pennation angle, but also neurophysiological adaptations such as increased motor-unit recruitment, increased motoneuron discharge rates and improved intermuscular coordination [60, 61]. The findings of this review indicate that greater increases in maximal strength as a response to periodized resistance training when compared to NP resistance training, and UP when compared to LP, cannot be attributed to morphological adaptations in the form of muscle hypertrophy, since the meta-analyses found no differences in measures of muscle hypertrophy between training protocols. While blunted muscle hypertrophy has been observed in older adults [62, 63], this finding cannot be attributed to the inclusion of older participants as this was accounted for by sensitivity analyses. Therefore, it seems reasonable to assume that variations in volume and intensity could induce greater neurophysiological adaptations, either at the spinal or supraspinal levels, possibly by increasing descending drive or by increasing α-motoneuronal net excitation by altering excitatory and/or inhibitory synaptic inputs leading to increased motor unit recruitment and increased discharge rates [64, 65], or alternatively through effects on coordination [66]. However, since none of the studies investigating effects of periodization have investigated potential differences in neurophysiological adaptations, the underlying mechanisms need to be investigated in future studies.

Although the results of the present meta-analyses investigating effects on muscle hypertrophy indicated that periodization does not affect muscle hypertrophy, current methods for measuring muscle hypertrophy may lack sensitivity to detect small differences [67]. Furthermore, the majority of studies included in the review used indirect measures to assess muscle hypertrophy, which are not considered the gold standard for assessing changes in human skeletal muscle mass [68]. A number of studies even used skinfold measurements to assess changes in FFM, a method that lacks precision compared to other indirect assessments of muscle mass, such as dual-energy X-ray absorptiometry (DXA) [69]. Therefore, separate analyses of direct and indirect measures of muscle hypertrophy could have been advantageous. However, since very few studies used direct measures to assess muscle hypertrophy, this approach was not feasible. Thus, it cannot be excluded that the varying degrees of accuracy and sensitivity between these methods may have contributed to the apparent lack of differences in muscle hypertrophy between training groups. It is also possible that subtle differences in muscle hypertrophy between NP and periodized resistance training are simply not detectable due to greater measurement error than difference in muscle hypertrophy, even with MRI [70], which is considered the gold standard for assessing changes in muscle mass. Therefore, it is conceivable that potential differences in muscle hypertrophy between training protocols may simply require longer training interventions to be detectable, and as such contribute to greater increases in strength over long periods of time.

4.2 Training Status

The results presented in this systematic review and meta-analysis demonstrated that regardless of resistance training status, periodized resistance training was superior at eliciting increases in maximal strength compared to NP resistance training. Thus, both untrained and trained individuals benefit from periodizing resistance training when the aim is to increase maximal strength. This may have practical applications for a wide range of individuals ranging from athletes to older adults and patients in rehabilitation settings. However, the possibility that the effect of periodization may be specific to the trained resistance exercises cannot be excluded. Therefore, the degree to which the increased strength, as measured by 1RM, would transfer to other activities, such as sports performance or activities of daily living is yet unknown.

Contrary to the results indicating that periodized resistance training has a greater effect on gains in maximal strength irrespective of training status, when compared to NP resistance training, a greater effect of UP in comparison to LP was observed only in trained participants. This indicates that (a) while some sort of periodization may be beneficial, daily or weekly undulations in volume and intensity may not be needed to maximize strength in untrained individuals, and (b) the need for frequent variations in volume and intensity may increase with training advancement, when the aim is to develop maximal strength. While this indicates that daily or weekly variations in volume and intensity are beneficial for increasing strength in resistance trained individuals, the reasons for this are unclear. However, as no differences in measures of muscle hypertrophy were found between LP and UP in neither trained nor untrained individuals, it can be speculated that more frequent variations in stimuli are beneficial in order to drive neurophysiological adaptations in trained individuals and thereby lead to increased strength. This is in accordance with previous literature suggesting that untrained individuals are capable of developing strength using basic resistance training programs, while trained lifters require greater variations in training stimuli and more advanced training strategies in order to maximize strength gains [12, 71].

Although several of the included studies were conducted on participants characterized as resistance trained, most of these studies characterized participants as resistance trained if they had > 6 months to > 1 year of experience with resistance training. This classification is often pragmatically appropriate due to logistics of participant recruitment; however, it also limits the applicability of the results in regard to highly trained athletes. This is especially important since periodization is intended to maximize performance in high-level competitive athletes. This limitation could potentially be solved by classifying trained individuals using a higher cut-off value of experience with resistance training. However, this would create the issue, that due to the classification of trained participants, many of the included studies had participants that may have had anywhere from 6 months to several years of resistance training experience. Thus, it would not be feasible to classify these participants as either trained or untrained, based on a specific cut-off value, due to a large degree of within-study variation in resistance training experience. Naturally, this may have impacted results.

4.3 How is Volume and Intensity Most Effectively Periodized When the Aim is Maximal Strength?

For the studies included in this review, resistance training protocols that used a periodized approach led – on average – to 21.6% greater increases in strength when compared to NP training. Therefore, it is recommended that practitioners use a periodized resistance training protocol to increase maximal strength. In addition, this review indicates that UP is superior to LP for increases in maximal strength. A major difference between these periodization models is the frequency of variation in volume and intensity, indicating that the difference in strength outcomes between LP and UP may be related to the frequency by which variations in volume and intensity occur. Based on the results, it seems that variations in volume and intensity should occur daily or weekly when the aim is to increase maximal dynamic strength. However, daily or weekly variations in volume and intensity, as is the case in UP, do not exclude the possibility of implementing other periodization models simultaneously, such as BP or LP. This demonstrates a valuable point, that different periodization models and programming strategies may be incorporated in a long-term training plan. In this manner, the simultaneous combination of different periodization models allows for more complex periodization strategies.

A study by Prestes et al. [72] indicated that a combination of UP and LP resulted in greater increases in 1RM compared to a combination of UP and RLP, suggesting that linear increases in intensity and decreases in volume over time seems to be superior to decreases in intensity and increases in volume over time. It has previously been suggested that DUP can be used to structure each week within a training plan using BP [73, 74]. As some blocks will be characterized by higher volumes and others by higher intensities, these blocks can be organized into a linear training plan, where blocks progress from high volume to high intensity over the course of a training program. Thus, illustrating an example of a training program utilizing a combination of DUP, BP, and LP. Based on this, it is speculated that implementing more complex periodization strategies in this manner, where frequent undulations in volume and intensity occur along with an increase in intensity and a decrease in volume over the course of a training program, may be superior at eliciting gains in maximal strength compared to more simple periodization models investigated in the current body of literature.

While the results of this review indicate that resistance training aimed at maximizing strength should be periodized, and that daily or weekly undulations in volume and intensity may be favorable for resistance trained individuals, there are several aspects of periodization and programming that cannot be inferred based on the literature. First, while meta-analyses allow isolation of a single variable within-studies, in this case periodization model, there are copious amounts of between-study variables that may affect the effects of periodization. These variables include, but are not limited to, exercise selection, intensity (i.e., %1RM or RM) and effort (i.e., how close to momentary muscular failure sets are performed). As such, it is possible that the observed effect of periodization on maximal strength may differ depending on the exercises trained, the range of intensities that are trained and how close to momentary muscular failure the resistance training is performed. An issue that has previously been discussed in the literature [75]. Secondly, there is high inter-individual variability in the response to resistance training [76]. This is important as the entire rationale for periodization is to maximize the response to training. As such, some individuals may respond better to one model of periodizing training, whereas another model of periodizing training may be more beneficial for others. Further, the individual response to a specific resistance training program likely changes over time. Thus, training protocols that are effective at eliciting increases in strength at one time point, may not be effective at a later time point for the same individual, requiring a change in training stimuli to continue to elicit the desired training adaptations. Therefore, the findings of this review should only be applied as a general recommendation for the planning of resistance training programs. Whereas at the individual level, monitoring the response to resistance training and adjusting variables based on the response, or lack thereof, likely is of greater importance for long-term strength-development.

This relates to another limitation of the current periodization literature, namely that of periodization being intended to maximize strength development in elite athletes with macrocycles of far greater duration than the study durations assessed in the literature. This is important, as previous studies have suggested that the benefits of periodized resistance training are greater in long-term training programs [28, 56, 59]. However, it remains unclear whether the effects of periodization increase or diminish over time. When examining individual studies, the study by Kraemer et al. [40] was the study of longest duration. This study reported that the larger increases in 1RM for the periodized training group in comparison to NP were greatest in the first 4 months of the training intervention and diminished over the course of the study. On the other hand, the study by De Souza et al. [32] reported no significant differences between NP and periodized resistance training during the first 6 weeks of the training intervention, whereas only periodized resistance training resulted in significant increases in 1RM squat and quadriceps muscle CSA the following 6 weeks, suggesting that the effects of periodization on strength gains and muscle hypertrophy may increase over time. As such, future research should address the question as to whether the effects of resistance training periodization increase or diminish over time, since this is beyond the scope of the present review and meta-analysis.

5 Conclusion

The findings of this systematic review and meta-analysis demonstrate that when comparing volume-equated resistance training programs in pre-post study designs, periodized resistance training has a greater effect on maximal strength compared to NP resistance training. Furthermore, daily and weekly undulations in volume and intensity are associated with greater increases in maximal strength, as indicated by a greater effect on 1RM in UP when compared to LP. However, subgroup analysis revealed that this was only the case for trained individuals, whereas no difference was found between LP and UP in untrained individuals. No differences were found in muscle hypertrophy between NP and periodized resistance training, along with no differences in muscle hypertrophy between different periodization models. These results suggest that periodization of resistance training does not affect muscle hypertrophy, when volume is equated between conditions. Based on this finding, we suggest that other factors such as neurophysiological adaptations may underpin the greater increases in strength observed with periodized resistance training and periodization models that involve daily or weekly undulations in volume and intensity.

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Declarations

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